

CS 70 — Discussion 2A

Problem 1: Leaves in a Tree

A leaf in a tree is a vertex with degree 1.

Part (a): Maximum number of leaves in a tree on $n \geq 3$ vertices

Claim: The maximum number of leaves is $m = n - 1$.

Achievability. Consider the star graph on n vertices: a central vertex c connected to each of the other $n - 1$ vertices. This graph has $n - 1$ edges and no cycles, so it is a tree. The central vertex has degree $n - 1$, and each of the other $n - 1$ vertices has degree 1, hence is a leaf. So the star has exactly $n - 1$ leaves.

Upper bound. The only integer greater than $n - 1$ is n , so it suffices to show that not all n vertices can be leaves. Suppose for contradiction that every vertex has degree 1. By the handshaking lemma,

$$\sum_v \deg(v) = 2|E|$$

The left side equals n , so $|E| = n/2$. But a tree on n vertices has exactly $n - 1$ edges, giving $n/2 = n - 1$, i.e., $n = 2$. This contradicts $n \geq 3$. Therefore at most $n - 1$ vertices can be leaves.

Hence the maximum is $m = n - 1$. □

Part (b): Every tree on $n \geq 2$ vertices has at least two leaves

Proof. Let P be a longest path in the tree, with endpoints a and b . Such a path exists because the tree is finite.

We claim a is a leaf. Suppose not, so $\deg(a) \geq 2$. Then a has a neighbor w other than its neighbor along P . There are two cases:

- If w is not on P , then prepending w to P yields a longer path, contradicting that P is longest.
- If w is on P , then the edge $\{a, w\}$ together with the portion of P between a and w forms a cycle, contradicting that the graph is a tree.

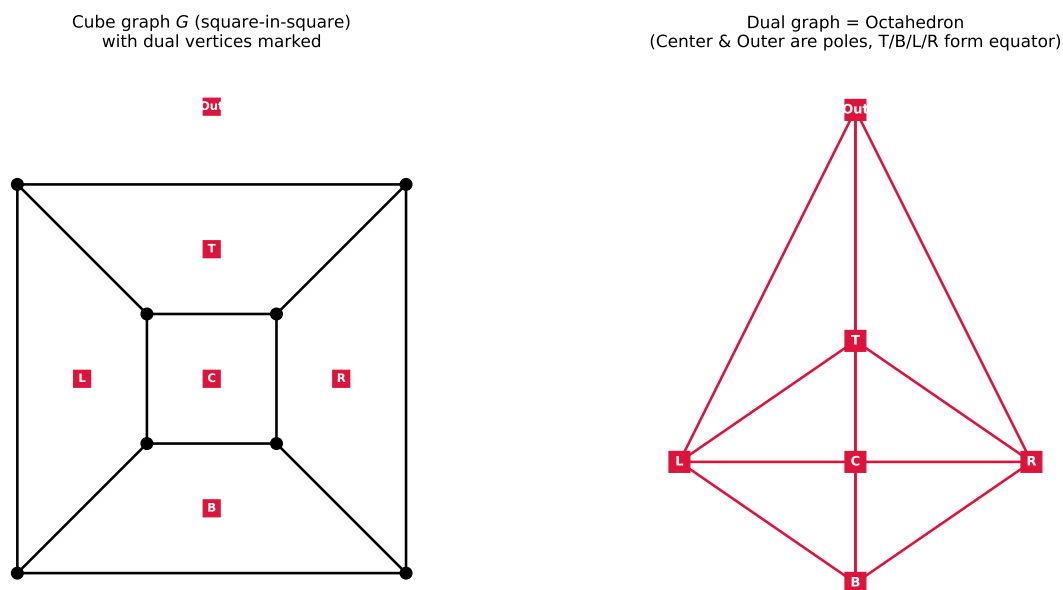
Both cases are contradictions, so $\deg(a) = 1$ and a is a leaf. By the identical argument, b is a leaf. Since P has $n \geq 2$ vertices, its endpoints a and b are distinct, giving two distinct leaves. □

Problem 2: Cube Dual

Part (a): Planar drawing of the cube graph and its dual

The cube graph G has 8 vertices (the corners of a cube) and 12 edges. It can be drawn in the plane as a “square inside a square,” with four spokes connecting corresponding corners.

The dual graph is formed by placing one vertex in each face of the planar drawing. The cube’s planar drawing has 6 faces: the inner square (C), the four regions between the squares (T, B, L, R), and the outer infinite face (Out). Two dual vertices are connected whenever their faces share an edge. The resulting dual is the octahedron.



Each vertex of the dual has degree 4: the center face C and outer face Out are the two “poles,” each connected to all four side faces T, B, L, R, which themselves form a 4-cycle (equator).

Part (b): Spanning tree and its dual correspondence

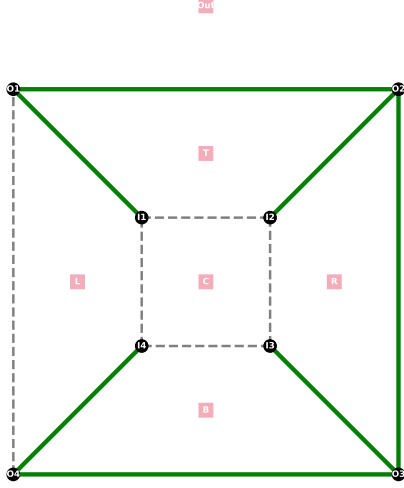
A spanning tree of G uses $8 - 1 = 7$ edges. We choose the outer path $O_1O_2O_3O_4$ together with the four spokes. This leaves out 5 edges: the fourth outer edge O_4O_1 and all four inner-square edges.

The duality relationship states: the edges left out of G ’s spanning tree correspond exactly to the edges of the dual’s spanning tree. Each left-out edge maps to the dual edge crossing it:

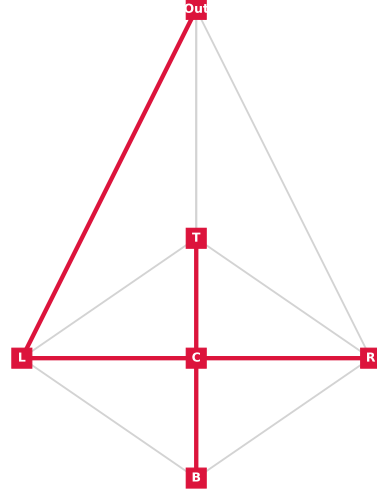
- Inner edges $I_1I_2, I_2I_3, I_3I_4, I_4I_1 \rightarrow$ dual edges C–T, C–R, C–B, C–L
- Outer edge $O_4O_1 \rightarrow$ dual edge L–Out

This gives the dual spanning tree $\{C-T, C-R, C-B, C-L, L-Out\}$, which has $6 - 1 = 5$ edges as required.

Spanning tree of G (green = tree, gray dashed = left-out)



Corresponding spanning tree of dual
 $\{C-T, C-R, C-B, C-L, L-Out\}$



Note that G 's spanning tree (7 edges) and the dual's spanning tree (5 edges) partition all 12 edges, and the two trees never cross.

Problem 3: Property T Implies Non-Planar

Property T: For every three distinct vertices v_1, v_2, v_3 , there are at least two edges among them.

Claim: If G has property T and $n \geq 7$ vertices, then G is non-planar.

Proof. A *non-edge* is a pair of vertices with no edge between them. Property T says every triple of vertices has at most one non-edge among its three pairs.

Each vertex is in at most one non-edge. Suppose a vertex v were in two non-edges, say $\{v, a\}$ and $\{v, b\}$ are both non-edges. Then the triple $\{v, a, b\}$ contains two non-edges, violating property T. So each vertex belongs to at most one non-edge.

Counting edges. Since each vertex is in at most one non-edge, the non-edges form a matching, so there are at most $\lfloor n/2 \rfloor$ of them. The total number of possible edges is $\binom{n}{2} = \frac{n(n-1)}{2}$, so the number of actual edges satisfies

$$e \geq \frac{n(n-1)}{2} - \left\lfloor \frac{n}{2} \right\rfloor \geq \frac{n(n-1)}{2} - \frac{n}{2} = \frac{n(n-2)}{2}.$$

Comparing to the planar bound. A simple planar graph satisfies $e \leq 3n - 6$. We show the lower bound above exceeds this for $n \geq 7$:

$$\frac{n(n-2)}{2} > 3n - 6 \iff n(n-2) > 6n - 12 \iff n^2 - 8n + 12 > 0 \iff (n-2)(n-6) > 0.$$

This holds for all $n \geq 7$. Therefore $e > 3n - 6$, which exceeds the maximum number of edges in a planar graph. Hence G is non-planar. $\square$

Problem 4: Planarity and Graph Complements

For a graph $G = (V, E)$, the complement $\overline{G} = (V, \overline{E})$ has the same vertices, and an edge exists in $\overline{G}$ if and only if it does not exist in G .

Part (a): Vertices and edges of $\overline{G}$

$\overline{G}$ has the same v vertices. The number of edges is the total number of possible pairs minus the edges of G :

$$\overline{e} = \binom{v}{2} - e = \frac{v(v-1)}{2} - e.$$

Part (b): For $v \geq 13$, G planar implies $\overline{G}$ non-planar

Proof. Suppose G is planar. By the planar edge bound, $e \leq 3v - 6$. Then the number of edges in $\overline{G}$ satisfies

$$\overline{e} = \frac{v(v-1)}{2} - e \geq \frac{v(v-1)}{2} - (3v-6).$$

We show this exceeds the planar bound $3v - 6$ for $\overline{G}$ (which has the same v vertices):

$$\frac{v(v-1)}{2} - (3v-6) > 3v-6 \iff \frac{v(v-1)}{2} > 6v-12 \iff v(v-1) > 12v-24 \iff v^2 - 13v + 24 > 0.$$

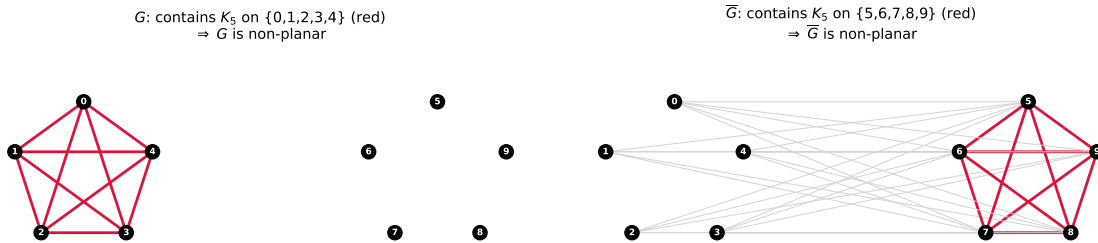
The roots of $v^2 - 13v + 24$ are $\frac{13 \pm \sqrt{73}}{2} \approx 2.23$ and 10.77 , so the quadratic is positive for $v \geq 11$, and in particular for all $v \geq 13$. Therefore $\overline{e} > 3v - 6$, so $\overline{G}$ exceeds the planar edge limit and is non-planar. $\square$

Part (c): Is the converse true?

The converse is false. We exhibit a graph G for which both G and $\overline{G}$ are non-planar, so that $\overline{G}$ non-planar does *not* imply G planar.

Construction. Take 10 vertices split into two groups $A = \{0, 1, 2, 3, 4\}$ and $B = \{5, 6, 7, 8, 9\}$. Let G consist of a complete graph K_5 on A (and no other edges). Then:

- G contains K_5 on A , so G is non-planar.
- In $\overline{G}$, the group B (which had no edges in G) becomes a complete K_5 , so $\overline{G}$ is non-planar.



Both G and $\overline{G}$ are non-planar, so the converse fails. $\square$